

could be introduced with a Java applet or by viewing apparently benign data that, in actuality, contains executable code. More so than ever, secure operating systems are needed to protect against this threat.

Here, we identify some features of secure operating systems that are necessary to protect application-space security mechanisms yet are lacking in mainstream operating systems. They form the ‘missing link’ of security. Although this section only deals with features, it is important to note that features alone are inadequate. Assurance evidence must be provided to demonstrate that the features meet the desired system security properties and to demonstrate that the features are implemented correctly. Assurance is the ultimate missing link; although approaches to providing assurance may be controversial, the importance of assurance is undeniable.

Mandatory Security

An operating system’s mandatory security policy may be divided into several kinds of policies, such as an access control policy, an authentication usage policy, and a cryptographic usage policy. A mandatory access control policy specifies how subjects may access objects under the control of the operating system. A mandatory authentication usage policy specifies what authentication mechanisms must be used to authenticate a principal to the system. A mandatory cryptographic usage policy specifies what cryptographic mechanisms must be used to protect data. Additionally, various subsystems of the operating system may have their own mechanism usage policies. These subsystem specific usage policies may be dependent on the cryptographic usage policy. For example, a network usage policy for a router might specify that sensitive network traffic should be protected using IPSEC ESP in tunnelling mode prior to being sent to an external network. The selection of a cryptographic algorithm for IPSEC ESP may be deferred to the cryptographic usage policy.

A secure system must provide a framework for defining the operating system’s mandatory security policy and translating it to